

MADHUMITA KOLUKULURI

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EDUCATION

Northeastern University, Boston, USA

Master of Science in Robotics with CS

Expected May 2026

CGPA: 3.75

Coursework: Robot Mechanics and Control, Reinforcement Learning, Computer/Human Interaction, Pattern Recognition and Computer Vision.

Malla Reddy Engineering College for Women, Hyderabad, India

Bachelor of Technology in Computer Science and Engineering

June 2024

CGPA: 3.45

Coursework: Data Structures and Algorithms, Big Data Analytics, Operating Systems, Machine Learning, Database Management Systems, DevOps on AWS, Peer-to-Peer protocols and local area networks.

SKILLS

- Technical Skills: Python, Java, SQL, C, HTML, MS Office, Power BI, Web Development, MATLAB, SIMULINK, TensorFlow
 - Soft Skills: Teamwork, Leadership, Time Management, Communication, Conflict Management
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EXPERIENCE

Northeastern University, Boston, USA (Teaching Assistant)

Jan 2025 – March 2025

- Evaluated student submissions on Reinforcement Learning concepts, while providing constructive feedback for improvement.
- Conducted office hours to assist students with course material, clarifying doubts related to algorithms used in the assignments.
- Provided structured guidance on problem-solving approaches and debugging RL implementations.

Sampradaya Vastra, Hyderabad, India (Consultant - MLE)

Feb 2024 – June 2024

- Developed a custom recommendation system using Matrix Factorization (SVD) and k-Nearest Neighbors (k-NN) algorithms, ultimately deploying a hybrid model that achieved a 25 percent improvement in precision @k and a 20 percent increase in recall, set to enhance user engagement upon the website's launch.
 - Leveraged Python libraries such as scikit-learn and TensorFlow to create a robust machine learning pipeline, implementing techniques like Grid Search for hyperparameter tuning, content moderation and Stratified K-Folds for cross-validation, ensuring an optimal balance between model complexity and accuracy.
 - Projected an increase in overall revenue by 18 percent upon implementation of the recommendation algorithm, demonstrating its potential to boost average order values and improve customer retention rates once the website is live.
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PROJECTS

AI Powered Research Assistant

January 2025 - April 2025

Northeastern University, Boston, USA

- Developed an AI-powered research assistant grounded in Human-Computer Interaction (HCI) principles, prioritizing user-centric design through Schneiderman's 8 golden rules and iterative feedback, enabling seamless paper summarization, citation retrieval, and intelligent comparison tailored for academic researchers.

Sorting Dice using Robotic Arm Control

January 2025 - April 2025

Northeastern University, Boston, USA

- Vision-guided robotic system using ROS and CNNs to autonomously classify and sort dice based on face values with precise motion planning.

Portfolio Management

September 2024- December 2024

Northeastern University, Boston, USA

- Optimized stock portfolios with Multi-Agent Reinforcement Learning, enhancing dynamic decision-making and risk management by integrating transaction costs and market volatility.